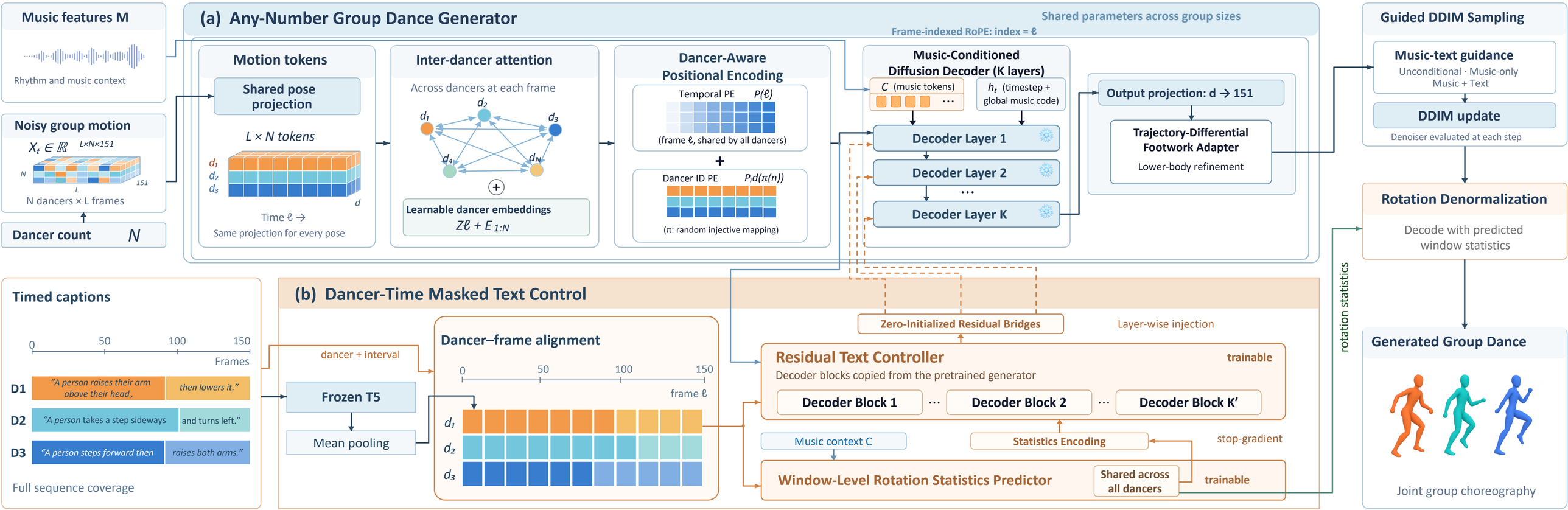

Music features M
Rhythm and music context
Noisy group motion
$X_t \in \mathbb{R}^{L\times N\times 151}$
N dancers × L frames
Dancer count N
(a) Any-Number Group Dance Generator
Shared parameters across group sizes
Frame-indexed RoPE: index = ℓ
Motion tokens
Shared pose projection
L × N tokens
Time ℓ →
Same projection for every pose
Inter-dancer attention
Across dancers at each frame
Learnable dancer embeddings
$Z\ell + E_{1:N}$
Dancer-Aware Positional Encoding
Temporal PE $P(\ell)$
(frame ℓ, shared by all dancers)
Dancer ID PE $P_i d(\pi(n))$
(π: random injective mapping)
Music-Conditioned Diffusion Decoder (K layers)
C (music tokens)
h_t (timestep + global music code)
Decoder Layer 1
Decoder Layer 2
Decoder Layer K
Output projection: d → 151
Trajectory-Differential Footwork Adapter
Lower-body refinement
Guided DDIM Sampling
Music-text guidance
Unconditional · Music-only
Music + Text
DDIM update
Denoiser evaluated at each step
Rotation Denormalization
Decode with predicted window statistics
Generated Group Dance
Joint group choreography
rotation statistics
Timed captions
Frames
D1 "A person raises their arm above their head," then lowers it."
D2 "A person takes a step sideways and turns left."
D3 "A person steps forward then raises both arms."
Full sequence coverage
(b) Dancer-Time Masked Text Control
dancer + interval
Frozen T5
Mean pooling
Dancer–frame alignment
frame ℓ
Zero-Initialized Residual Bridges
Layer-wise injection
Residual Text Controller
trainable
Decoder blocks copied from the pretrained generator
Decoder Block 1
Decoder Block 2
Decoder Block K'
Music context C
Statistics Encoding
stop-gradient
Window-Level Rotation Statistics Predictor
Shared across all dancers
trainable